RESEARCH ARTICLE

When resolution consistency fails: an information-matched audit for disaster remote sensing

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ARTICLE HISTORY
Compiled July 25, 2026

Running head: Auditing resolution reliability

ABSTRACT
Reliability scores under spatial-resolution shift are often compared without checking whether they receive the same information. We audit a multi-scale Jensen–Shannon consistency score that ranks errors on an eight-fold degraded image while retaining access to its unavailable finer original. Exact-hash- clean AIDER and Hurricane Damage splits, three independently trained ResNet-18 seeds per corpus, and matched confidence, energy, Earth-observation nearest- neighbour, virtual-logit matching and received-image consistency baselines are used. Removing the finer reference reduces mean consistency error area under the receiver operating characteristic curve (AUROC) from 0.964 to 0.623 on AIDER and from 0.989 to 0.590 on Hurricane Damage. The reference-dependent minus received-image AUROC gaps are 0.341 and 0.399, respectively; all six seed-level 10,000-replicate paired bootstrap intervals exclude zero. No deployable score dominates across corpora and seeds, and native-calibrated thresholds do not preserve both coverage and low selective risk. We then replace overlapping geospatial crops with guarded blocks and evaluate 1,441 same-building UAS/post-event-satellite pairs from four acquisitions at three geographies and two hurricane events. The intersection-controlled UAS classifier exceeds the majority baseline for every seed. Received-image consistency has a positive pooled satellite AUROC lift over confidence, but the effect reverses at both Hurricane Michael acquisitions; spatial-cluster intervals include zero. The mean-based threshold-transfer criterion passes, but its post-hoc all-seed robustness sensitivity fails. The findings show a large oracle–received protocol gap and demonstrate that acquisition pooling can reverse operational conclusions, while exact duplicates and spatial overlap require explicit controls. We develop an information-set and leakage audit protocol for resolution-reliability studies.

1. Introduction

Spatial resolution varies across uncrewed-aerial-system and satellite disaster products because altitude, optics, acquisition geometry, mosaicking and delivery constraints vary. A classifier can consequently encounter a coarse image while remaining confident in an incorrect prediction. Selective prediction and out-of-distribution (OOD) scores are intended to identify such unreliable cases, but their evaluation is meaningful only when competing scores receive the same information.

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Multi-view disagreement is a plausible reliability signal. Predictive- distribution divergences can forecast performance under distribution shift (Schirmer, Zhang, and Nalisnick 2023), and Jensen–Shannon (JS) consistency has an established role in corruption-robust training (Hendrycks et al. 2020). However, a synthetic resolution ladder creates a specific failure mode. When an error made on an eight-fold degraded image is scored by disagreement with the retained undegraded original, the score receives a finer reference that a system receiving only the coarse image does not possess. Comparing it with confidence or feature distance from the coarse image conflates score quality with information access.

Evaluation data can introduce a second form of optimism. Exact images can occur across nominal splits, and fixed crops from nearby geospatial objects can share source pixels even when object identifiers differ. Duplicate contamination is known to bias image benchmarks (Barz and Denzler 2020), while spatial dependence can substantially inflate remote-sensing accuracy estimates (Karasiak et al. 2022, Kattenborn et al. 2022). These issues are especially important for reliability studies because they change both classifier errors and the apparent ability to rank those errors.

This article presents a pre-specified, failure-finding audit of reliability under resolution shift. It makes four contributions:

(1) It defines an information-set audit that contrasts reference-dependent and received-image consistency for the same coarse-image errors.
(2) It compares maximum-softmax confidence, maximum logit, energy, Earth- observation (EO) deep nearest neighbours, virtual-logit matching (ViM), and received-image consistency under identical source-only information.
(3) It replaces contaminated evaluations with encoded-byte duplicate exclusion and guarded spatial blocks whose retained 512-pixel crops have zero cross-split intersections.
(4) It extends measured-ground-sampling-distance (GSD) evaluation to paired UAS and post-event satellite buildings from four held-out acquisitions at three geographies and two disaster events, with independent model seeds and paired spatial-cluster uncertainty.

The intended outcome is deployment guidance and an executable audit protocol, not a claim that JS disagreement or any existing OOD score is a new primitive. Negative and superseded results are retained as part of the evidence.

2. Related work

2.1. Disagreement and reliability

Predictive disagreement is widely used as a proxy for epistemic uncertainty. Schirmer, Zhang, and Nalisnick (2023) show that full-distribution divergences, including JS divergence, can be more informative than top-class agreement for performance forecasting under shift. AugMix instead uses JS consistency as a training objective across augmentations (Hendrycks et al. 2020). These precedents mean that neither JS divergence nor transformation consistency is itself the novelty of this study. The present question is whether a divergence-based score has been evaluated with information unavailable at deployment.

Our verified literature set does not establish that this exact unavailable- fine-reference comparison is prevalent in published EO practice. The failure mode was exposed in a prior internal evaluation and is studied here as a concrete counterexample and audit case, not as an estimate of field-wide prevalence.

2.2. OOD detection in Earth observation

Remote-sensing OOD work considers unseen classes, sensors and geographical domains. The remote-sensing Dirichlet prior network (DPN-RS) uses in-domain and auxiliary OOD examples to learn distributional uncertainty (Gawlikowski et al. 2022). Post-hoc alternatives avoid additional OOD training. Deep nearest neighbours use distance to normalised source features (Sun et al. 2022); ViM combines a principal-space feature residual with classifier logits (Wang et al. 2022). A ten-method EO benchmark found ViM and nearest neighbours among the strongest methods without retraining (Li et al. 2024). We therefore include both, but distinguish prediction-error detection from conventional binary in-/out-of-distribution classification.

Target-assisted EO detectors answer a different information-budget question. TARDIS uses known and wild-domain samples to construct an auxiliary shift detector (Ekim et al. 2025); allowing the evaluation pool as its wild set would be transductive relative to our source-only scores. EarthShift supplies real-world EO shift tasks, including spatial-resolution variation, rather than a directly interchangeable per-image error score (Doerksen and Kerner 2026). We therefore treat these as adjacent protocols and compare the strongest source-only post-hoc methods under a common received-image budget.

2.3. Leakage and spatial validation

Near-duplicate removal changes measured performance on established image benchmarks (Barz and Denzler 2020). Geospatial collections can contain even more severe split duplication (Adimoolam, Poullis, and Averkiou 2026). Separately, nearby remote-sensing observations are spatially dependent. Buffered or spatial leave-one-out validation is needed when the intended inference concerns new areas (Karasiak et al. 2022), and random hold-outs can overstate convolutional-network performance (Kattenborn et al. 2022). Our audit checks both encoded-image identity and source-pixel intersection between fixed crops; these are complementary failure modes.

3. Audit method

3.1. Information sets

Let f be a classifier and $\boldsymbol{p}_s(\mathbf{x})$ its predictive distribution after downsampling an original image by factor s and resizing it to the network input. The evaluated coarse prediction is $\hat{y}_8 = \arg\max_c p_{8,c}(\mathbf{x})$.

The reference-dependent consistency score is

$$D_{\mathrm{ref}}(\mathbf{x}) = \frac{1}{3} \sum_{s \in \{2,4,8\}} \mathrm{JS}(\boldsymbol{p}_1(\mathbf{x}), \boldsymbol{p}_s(\mathbf{x})) \, . \tag{1}$$

It uses the retained $s = 1$ image while ranking errors made at $s = 8$. Accordingly, we label it a privileged diagnostic.

For the deployable counterpart, let $\mathbf{z}$ be the image actually received at effective scale 8. Only further degradations of $\mathbf{z}$ are available:

$$D_{\mathrm{recv}}(\mathbf{z}) = \frac{1}{3} \sum_{r\in\{2,4,8\}} \mathrm{JS}(\boldsymbol{p}_1(\mathbf{z}), \boldsymbol{p}_r(\mathbf{z}))\,, \tag{2}$$

which corresponds to effective original scales $8, 16, 32, 64$. Both scores rank the same errors of $\boldsymbol{p}_1(\mathbf{z})$. Their paired error-detection AUROC difference,

$$\Delta_{\mathrm{oracle}} = \mathrm{AUROC}(D_{\mathrm{ref}}) - \mathrm{AUROC}(D_{\mathrm{recv}}), \tag{3}$$

is the descriptive oracle–received protocol gap rather than a method improvement. It combines finer-reference access with a change in the absolute degradation regime. A post-hoc mechanism sensitivity therefore anchors both scores on the received $s = 8$ prediction, uses three comparisons in each score, and contrasts finer views $\{1, 2, 4\}$ with coarser views $\{16, 32, 64\}$. This sensitivity controls the anchor and comparison count but still changes degradation direction; it is not interpreted causally.

3.2. Matched deployable scores

Every deployable score receives $\mathbf{z}$, source training data and source validation data only. We evaluate one minus maximum softmax probability, negative maximum logit, energy, deep nearest-neighbour distance, ViM (Wang et al. 2022), and D_{recv}. The nearest-neighbour bank uses normalised penultimate source features (Sun et al. 2022); k is selected from $\{1, 5, 10, 20, 50\}$ by source-validation error AUROC. Let W and $\mathbf{b}$ denote the final linear classifier's weight matrix and bias vector, and let W^+ denote the Moore–Penrose pseudoinverse. ViM constructs the classifier origin $-W^+\mathbf{b}$, estimates a principal source-feature subspace, and appends the scaled residual norm as a virtual OOD logit. Its principal dimension is selected from $\{64, 128, 256, 384\}$ using source validation only. Both selections use labelled source-validation images degraded to the target synthetic factor $s = 8$. This deliberately favourable, shift-specific supervised tuning is reported separately from untuned confidence, max-logit and energy scores.

3.3. Leakage controls

For AIDER and Hurricane Damage, we compute SHA-256 over encoded image bytes. Every member of a cross-split hash group is excluded; conflicting-label groups are reported explicitly. All other assignments remain fixed.

For CRASAR-U-DROIDs, 512-by-512 crops are centred on labelled buildings. Source-image coordinates are partitioned into 2048-by-2048 blocks assigned deterministically to training, validation or internal test sets. A crop is retained only when its centroid lies at least 256 pixels inside every block boundary. We then exhaustively test axis-aligned crop rectangles and require zero intersections between different splits.

3.4. Paired measured-GSD audit

The measured-GSD evaluation pairs identical building identifiers with matching eligible labels in UAS and post-event satellite products. Reliability is computed separately from each received crop; no UAS view is supplied when scoring its satellite pair. Thresholds at target coverages 0.5, 0.7 and 0.9 are calibrated on guarded source validation and transferred unchanged. We report realised coverage, absolute coverage error, selective risk and false-critical rate.

All GSD values are read from the fixed CRASAR `statistics.csv` at dataset revision `47cf4ab3a94d42978975f7d23338a996125ac0e9`; the captured metadata file has SHA-256 `b2a2fee6b1a631e2f2dc5f24a98988f3152728fd5dd793fc62b7707271a38936`.

Uncertainty is estimated with 10,000 paired spatial-cluster bootstrap replicates. Sites are top-level clusters and 2048-by-2048 UAS source blocks are resampled within site, retaining every paired building in a sampled block. A post-hoc dependence sensitivity additionally groups pairs into connected components whenever either their UAS or satellite 512-pixel crop rectangles overlap, and reports site-specific intervals. This stronger clustering is not used to redefine the preregistered criteria.

4. Experimental setup

4.1. Datasets

AIDER version 1.0 (CC BY 4.0) supplies five disaster-scene classes and the primary synthetic resolution audit (Kyrkou 2020). Its corrected train/validation/test counts are 4,499/961/969. Satellite Images of Hurricane Damage (CC BY 4.0) supplies an independent binary corpus through the Hugging Face revision `4bb60de01b2d9e323d364d76d8b08c2eaeef1a64`; corrected counts are 6,997/997/2,000 (Cao and Choe 2018). CRASAR-U-DROIDs (CC BY 4.0) provides source coordinates, building-damage labels, measured GSD metadata, and aligned UAS/satellite products at revision `47cf4ab3a94d42978975f7d23338a996125ac0e9` (Manzini et al. 2024).

The AIDER frozen split is corrected only by excluding every member of the two conflicting-label exact-duplicate groups discovered by the pre-specified hash audit. The Hurricane split uses the fixed seed-0 70/10/20 index assignment and the same all-members exclusion rule for cross-split hashes.

Intersection-controlled CRASAR classifier training uses guarded crops from Lancaster Canyon Gate (Hurricane Harvey) and Summerlin San Carlos (Hurricane Ian). The held-out paired evaluation uses Harlem Heights and McGregor College Parkway South 1 (Hurricane Ian), plus the 13 and 14 October Mexico Beach products (Hurricane Michael). Evaluation sites do not contribute images, labels or outcomes to training or model selection.

4.2. Training

The primary classifier is the ImageNet-1K-initialised `microsoft/resnet-18` checkpoint at revision `65a5785d9156231087c481e0c7dd33a5ff6f7e3e`. We train independent seeds 101, 202 and 303 with AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}), batch size 64, and class-weighted cross-entropy. For class c, the loss weight is $N/(Cn_c)$, where N is the training-set size, C the number of classes, and n_c the class count. The checkpoint with the highest source-validation macro-F1 is retained. AIDER uses six epochs, Hurricane Damage five, and intersection-controlled CRASAR 12.

Training first resizes to 256 pixels, then applies a random 224-pixel resized crop with scale range 0.75–1.0 and a horizontal flip with probability 0.5. Evaluation uses resize-to-256 and a 224-pixel centre crop. Both use ImageNet channel normalisation.

Table 1. Encoded-byte cross-split duplicate audit. Every member of each listed group was excluded before retraining.

Dataset	Groups	Excluded	Retained
AIDER	2	4	6429
Hurricane Damage	3	6	9994

Synthetic resolution degradation uses bicubic downsampling and upsampling before these transforms. Runs use Python 3.10.20, PyTorch 2.12.1, torchvision 0.27.1, Transformers 5.12.1 and Apple M4 Max Metal Performance Shaders.

4.3. Metrics and fixed analysis

Error detection is evaluated with AUROC, area under the precision–recall curve (AUPRC), and false-positive rate at 95% error recall (FPR95). Classification metrics include accuracy, balanced accuracy and macro-F1. Threshold-transfer metrics are coverage, absolute target-coverage error, selective risk and false-critical rate.

Paired score differences use 10,000 within-seed bootstrap replicates. The measured-GSD analysis additionally uses the paired spatial-cluster bootstrap defined above. Seed-level values and sample standard deviations are reported; model-seed variation is not substituted for building- or site-sampling uncertainty.

4.4. Pre-specified criteria

Criterion A1 requires positive privileged-minus-received mean AUROC on both synthetic corpora, a positive paired bootstrap interval for every seed, and six positive seed differences. W11 requires pooled held-out UAS accuracy above the pooled majority baseline for all seeds and mean balanced accuracy of at least 0.60. W12 requires positive mean received-consistency lift on satellite views, positive seed-specific spatial-cluster intervals, and positive mean lift at three or more of four sites. W13 requires the mean satellite absolute coverage error to be no larger for received consistency than confidence and the mean received-consistency selective risk to be no higher at target coverage 0.70. Requiring both inequalities for every seed is reported only as a post-hoc robustness sensitivity.

5. Results

5.1. Duplicate audits and corrected training

The encoded-byte audit found two cross-split AIDER hash groups. Both had conflicting labels; excluding every member removed four images and left 6,429. The Hurricane Damage audit found three cross-split groups with matching labels; the all-members rule removed six images and left 9,994. No encoded-byte hash remained in more than one split (Table 1).

Table 2. Paired oracle–received gap in error AUROC. Brackets give 95% paired bootstrap intervals from 10,000 replicates.

Seed	AIDER	Hurricane Damage
101	0.351 [0.312, 0.391]	0.354 [0.330, 0.378]
202	0.216 [0.183, 0.249]	0.439 [0.413, 0.465]
303	0.455 [0.415, 0.495]	0.405 [0.380, 0.430]

Table 3. Post-hoc anchor-matched mechanism sensitivity. Both scores anchor on the received $s = 8$ prediction and use three comparisons; the table reports fine-reference minus further-degradation error AUROC with 95% paired bootstrap intervals.

Seed	AIDER	Hurricane Damage
101	0.346 [0.304, 0.387]	0.351 [0.327, 0.375]
202	0.209 [0.175, 0.243]	0.439 [0.413, 0.465]
303	0.452 [0.411, 0.493]	0.404 [0.379, 0.429]

5.2. Oracle–received protocol gap

The reference-dependent diagnostic and received-image consistency ranked the same errors at effective scale 8 but differed in access to finer views. On AIDER, their mean AUROCs were 0.964 ± 0.013 and 0.623 ± 0.113, respectively. The mean oracle–received AUROC gap was 0.341 ± 0.120. On Hurricane Damage, mean AUROC changed from 0.989 ± 0.003 to 0.590 ± 0.045, a gap of 0.399 ± 0.043.

All six seed-level differences were positive. Every 10,000-replicate paired bootstrap interval excluded zero: AIDER intervals ranged from $[0.183, 0.249]$ to $[0.415, 0.495]$, and Hurricane intervals ranged from $[0.330, 0.378]$ to $[0.413, 0.465]$ (Table 2). The prespecified A1 criterion therefore passed.

Because the two primary protocols also differ in anchor and absolute degradation regime, we ran a post-hoc mechanism sensitivity that anchors both scores on $s = 8$ and uses three comparisons. Replacing the unavailable finer views $\{1, 2, 4\}$ with further degradations $\{16, 32, 64\}$ produced mean AUROC gaps of 0.336 ± 0.122 on AIDER and 0.398 ± 0.044 on Hurricane Damage; all six paired intervals remained positive (Table 3). This controls anchor and comparison count, but not degradation direction, and is reported as sensitivity evidence rather than a new confirmatory endpoint.

5.3. Matched deployable baselines

Under matched received-image information, no consistency score dominated the post-hoc baselines (Table 4). ViM had the highest mean AIDER AUROC (0.758), closely followed by confidence (0.754); received-image consistency reached 0.623. On Hurricane Damage, ViM reached 0.643, EO-kNN 0.618, received-image consistency 0.590, and confidence 0.542. These rankings varied by seed: received-image consistency exceeded confidence strongly for Hurricane seed 101, was lower for seed 202, and was statistically indistinguishable for seed 303. The evidence therefore supports the oracle–received gap, not universal superiority of one deployable score.

The preregistered paired ViM-minus-EO-kNN comparison was recomputed from

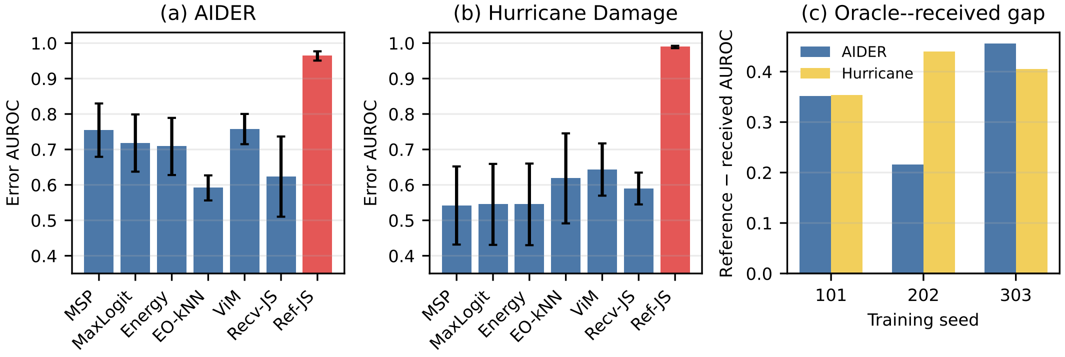

Figure 1. Information-matched error detection. Bars in the first two panels are mean AUROC across independent training seeds; error bars are sample SD. The red bar is a non-deployable diagnostic with access to the retained finer reference. The third panel shows its paired AUROC gap over received-image consistency for every seed. MSP denotes maximum softmax probability; Recv-JS and Ref-JS denote received-image and reference-dependent consistency.

Table 4. Error-detection AUROC under matched received-image information. The final row is a non-deployable diagnostic with access to the retained finer reference. Values are mean ± sample SD across three independently trained seeds.

Score	AIDER	Hurricane Damage
Confidence	0.754 ± 0.075	0.542 ± 0.110
Max logit	0.718 ± 0.081	0.545 ± 0.114
Energy	0.708 ± 0.081	0.545 ± 0.115
EO-kNN	0.591 ± 0.035	0.618 ± 0.127
ViM	0.758 ± 0.043	0.643 ± 0.074
Received consistency	0.623 ± 0.113	0.590 ± 0.045
Reference-dependent diagnostic	0.964 ± 0.013	0.989 ± 0.003

per-example arrays. All three AIDER intervals were positive (0.132 or greater at the lower bound). On Hurricane Damage, seed 101 was positive, seed 202 was negative, and seed 303 included zero. ViM therefore did not dominate EO-kNN across training runs.

5.4. Threshold transfer

Native-validation thresholds did not preserve both target coverage and low selective risk under severe shift. At target coverage 0.70 on AIDER, mean realised coverage was 0.384 for confidence, 0.257 for ViM, and 0.228 for received-image consistency. ViM selected a lower-risk subset (mean selective risk 0.040) but missed the target by 0.443 on average.

On Hurricane Damage, confidence, EO-kNN and ViM mean coverages were 0.029, 0.003 and below 0.001. Received-image consistency instead accepted 0.974 on average, but its mean selective risk was 0.467. Thus high transferred coverage did not indicate safe auto-decision retention. Reporting coverage without risk would reverse the operational interpretation.

Table 5. Intersection-controlled CRASAR classifier and pooled held-out paired results. Values are mean ± sample SD across seeds.

Endpoint	Value
Guarded internal accuracy	0.794 ± 0.007
Guarded internal majority accuracy	0.635
UAS accuracy	0.747 ± 0.014
Satellite accuracy	0.486 ± 0.081
Satellite confidence AUROC	0.465 ± 0.067
Satellite received-consistency AUROC	0.614 ± 0.079

Table 6. Held-out same-building measured-GSD evaluation. GSD is reported in $\mathrm{cm\,px^{-1}}$. The final column is the mean received-consistency minus confidence error-AUROC difference across three seeds.

Acquisition	Event	Pairs	UAS GSD	Satellite GSD	AUROC lift
Harlem Heights	Hurricane Ian	531	4.67	30.52	0.143
Mcgregor College Parkway South 1	Hurricane Ian	269	3.84	30.52	0.159
Mexico Beach 2018 10 13	Hurricane Michael	337	1.96	58.97	-0.155
Mexico Beach 2018 10 14	Hurricane Michael	304	1.95	58.97	-0.073
Pooled	Two events	1441	–	–	0.149

5.5. *Intersection-controlled measured-GSD replication*

The guarded CRASAR split retained 1,628 crops from two training sites: 975 for training, 267 for validation and 386 for internal testing. Exhaustive rectangle comparison found zero cross-split 512-pixel intersections. Internal test accuracy was 0.794 ± 0.007, above the 0.635 majority baseline for every seed; balanced accuracy was 0.781 ± 0.013.

The held-out paired evaluation contained 1,441 buildings in 708 UAS spatial blocks across four acquisitions at three geographies and two hurricane events (Table 6). Same-eligible-label selection retained 531/978 common identifiers at Harlem, 269/366 at McGregor, 337/507 at Mexico Beach 13 October, and 304/634 at Mexico Beach 14 October. These exclusions are reported because the resulting sample conditions on cross-product label agreement. Pooled UAS accuracy was 0.747 ± 0.014, above the 0.533 pooled majority baseline for every seed, and balanced accuracy was 0.752 ± 0.014. Criterion W11 therefore passed. Satellite accuracy fell to 0.486 ± 0.081, below the same majority baseline on average, confirming a severe cross-platform shift without invalidating the UAS classifier.

The post-hoc joint-overlap sensitivity connected pairs whenever either their UAS or satellite crops overlapped. It produced two effective clusters at Harlem and one at each remaining site because the larger satellite footprints overlap extensively. Pooled intervals continued to include zero, while site-specific cluster intervals were degenerate and are not interpreted as independent site inference.

5.6. *Measured-GSD ranking and threshold transfer*

On pooled satellite views, received-image consistency reached error AUROC 0.614 ± 0.079, compared with 0.465 ± 0.067 for confidence; all three seed differences were

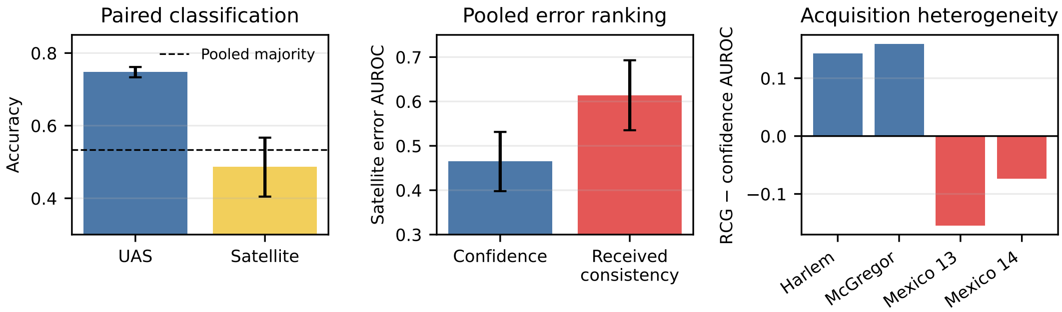

Figure 2. Intersection-controlled paired measured-GSD audit. Error bars in the first two panels are sample SD across training seeds. The acquisition panel shows that the pooled received-consistency advantage is confined to the two Hurricane Ian sites and reverses at both held-out Hurricane Michael sites.

positive and the mean difference was 0.149. This pooled summary did not generalise by site. Mean differences were 0.143 at Harlem Heights and 0.159 at McGregor, but -0.155 and -0.073 on the two Mexico Beach sites. The paired spatial-cluster bootstrap intervals for the pooled difference included zero for every seed. W12 therefore failed.

Measured threshold transfer was also heterogeneous. At target coverage 0.70, received-image consistency achieved mean satellite coverage 0.701 ± 0.138, compared with 0.499 ± 0.068 for confidence. Mean selective risk was lower (0.456 versus 0.548), but seed 202 reversed this ordering (0.549 versus 0.530), and spatial-cluster risk-difference intervals included zero. The preregistered mean-based W13 therefore passed, while the stricter post-hoc all-seed robustness sensitivity failed. The expanded evaluation changes the conclusion from a positive two-site ranking result to event-associated heterogeneity with no confirmatory ranking success and no seed-uniform threshold-transfer robustness.

6. Discussion

The audit distinguishes a useful diagnostic question from a deployable reliability claim. A reference-dependent score asks whether the prediction on a coarse synthetic view disagrees with a finer image that the benchmark retains. This can diagnose resolution sensitivity when both views genuinely exist. It does not establish that a system receiving only the coarse image can reproduce the same ranking or threshold-transfer behaviour.

Information matching changes the interpretation of baseline comparisons. Confidence, energy, nearest-neighbour distance, virtual-logit matching, and received-image consistency can all be computed from the received image plus source data. Their relative performance therefore concerns the score rather than hidden input access. A reference-dependent score is retained as an explicit upper-information diagnostic, not presented as an operational competitor.

The data audits have a related purpose. Exact duplicate checks target literal reuse across splits, whereas guarded spatial blocks target shared pixels and local spatial dependence between fixed crops. Neither control alone establishes geographical generalisation. The held-out site and event design therefore complements, rather than replaces, the lower-level leakage checks.

The multi-event result illustrates why pooling is insufficient. Received-image consistency had a positive pooled satellite AUROC difference because the two Hurricane Ian acquisitions favoured it, while both Hurricane Michael acquisitions favoured confidence. Cluster intervals consequently included zero despite three positive pooled seed differences. Likewise, mean threshold-transfer coverage and risk favoured consistency, so preregistered W13 passed, but one seed reversed the risk ordering and the post-hoc all-seed sensitivity failed. Consequently, W12 does not support general ranking and W13 does not establish seed-uniform robustness; the measured-GSD evidence supports event-associated heterogeneity and audit practice rather than a winning gate.

7. Limitations and threats to validity

The synthetic ladder is a controlled image transformation, not a physical sensor model. It isolates information access but does not recreate atmospheric, optical or processing effects associated with a changed GSD. The paired CRASAR comparison adds measured GSD but changes platform, sensor, view geometry, processing and physical footprint together; it therefore supports a cross-platform resolution-shift conclusion rather than a causal effect of GSD alone.

The primary reference-dependent and received-image scores also differ in absolute scale regime. Their AUROC difference is therefore a descriptive oracle–received protocol gap, not an identified causal effect of reference access alone. The anchor-matched sensitivity controls the anchor and comparison count but still contrasts finer with coarser degradation directions.

The audit evaluates disaster scene and binary building-damage classifiers with one backbone family. Three independent seeds quantify training variation but do not establish architecture universality. The held-out measured-GSD sites cover two hurricane events, not every hazard or geographical region. The two Mexico Beach acquisitions share geography and the effective event-level sample size is two; event-associated sign differences are descriptive rather than event-general inference.

Paired buildings are restricted to identifiers with the same eligible label in both products. This avoids label disagreement as an explanation for prediction changes but conditions the sample on cross-product agreement; the resulting class proportions are not population prevalence estimates. The protocol files, local execution logs and hashes record that criteria were frozen before their corresponding evaluations, but they were not deposited in an external preregistration registry before execution. Independent external timestamp verification is therefore unavailable.

ViM and nearest neighbours are strong source-only post-hoc baselines in EO benchmarks, but methods trained with auxiliary OOD data answer a different information-budget question. DPN-RS is consequently discussed rather than mixed into the matched source-only ranking. Finally, exact hashes do not detect all near-duplicates; the public release retains encoded hashes and feature outputs so a future perceptual audit can extend the analysis.

8. Conclusion

Reliability comparisons under resolution shift require an explicit account of what each score can observe. This study separates a reference-dependent resolution diagnostic from scores that can be computed when only the received image is available, then combines that information audit with exact-duplicate, spatial-intersection and measured-

GSD checks. The resulting protocol is designed to prevent an unavailable finer view or shared source pixels from being mistaken for a stronger uncertainty method. Its practical recommendation is simple: declare the inference-time information set, match it across baselines, audit image identity and crop geometry, and retain failed threshold-transfer results before making an operational reliability claim.

Acknowledgements

During preparation of this work, the authors used the Cursor artificial-intelligence assistant with GPT-5.6 Sol for software-development assistance, reference checking, result verification, and language refinement. The authors reviewed the source and revised versions, checked all quantitative claims against the registered execution artefacts, and take full responsibility for the final manuscript. No artificial-intelligence tool is listed as an author, and no artificial-intelligence-generated image is included.

Funding

No specific funding was received for this work.

Disclosure statement

No competing interest was reported by the authors.

Data availability statement

AIDER version 1.0 is available from Zenodo under CC BY 4.0 at `https://doi.org/10.5281/zenodo.3888300` (Kyrkou 2020). Satellite Images of Hurricane Damage is available from IEEE DataPort and its documented Hugging Face mirror under CC BY 4.0 at `https://doi.org/10.21227/sdad-1e56` (Cao and Choe 2018); this study fixes mirror revision `4bb60de01b2d9e323d364d76d8b08c2eaeef1a64`. CRASAR-U-DROIDs version 1 is available from DesignSafe-CI and Hugging Face under CC BY 4.0 at `https://doi.org/10.17603/ds2-fqy5-n314` (Manzini et al. 2024); this study fixes revision `47cf4ab3a94d42978975f7d23338a996125ac0e9`. No third-party imagery is redistributed. Version 2.0.2 of the complete code and evidence package is available at `https://github.com/akssha74/rcg-tt` and archived under the version-independent DOI `https://doi.org/10.5281/zenodo.21510152` (Sharma and Prasad 2026). The archive contains exact split files, portable per-example scores, source manifests, checkpoints, execution logs, claim and citation ledgers, and generated tables and figures. Code is MIT-licensed and generated research artefacts are CC BY 4.0.

Appendix A. Reproducibility and audit details

The supplementary release records the encoded-byte hashes used for duplicate audits, every excluded item, guarded-block coordinates, all retained crop rectangles, and the exhaustive cross-split intersection result. Seed-level checkpoints and raw score arrays are retained so that AUROC, AUPRC, FPR95, threshold-transfer, and bootstrap calculations can be independently recomputed.

Compressed per-example arrays contain labels, predictions and all synthetic audit scores for every corpus/seed. The paired measured-GSD arrays additionally contain UAS and satellite predictions, confidence, received consistency, site, event, UAS block, satellite block and joint overlap-cluster identifiers. Portable indexes omit local filesystem paths. The reproduction command verifies their hashes, recomputes headline metrics and the preregistered W13 adjudication, regenerates publication artefacts and clean-builds the paper.

Failed and superseded branches remain part of the release. In particular, the initial unpaired measured-GSD criterion, the overlapping-crop analysis, the reference-dependent operational interpretation, and the failed received-image-only W8/W9 experiment are not deleted when corrected evidence is generated.

The first implementation applied an all-seed W13 rule although the frozen criterion was mean-based. The corrected adjudication records W13 as passed and the all-seed requirement as a failed post-hoc robustness sensitivity. Other independent-review sensitivities, including anchor-matched finer-view scoring and joint UAS/satellite overlap clustering, are explicitly labelled post hoc.

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